Enterprise Employee Analytics — Detailed Data Flow Diagram (DFD)

End-to-End Pipeline: Raw Kaggle Ingestion -> SCD2 Warehouse -> Executive Streamlit Views

Step 1

Data Synthesis & Ingestion

- 1,470 Kaggle seed records
- Faker scales to 100k+ employees
- Generates historical career trails
- Injects promotion & dept changes
- Saves stg_ibm_hr_raw.csv

Staging Load

Step 2

Staging DB Storage

- Database: enterprise_staging
- Table: stg_ibm_hr_raw
- Preserves raw string fields
- Indexed on EmployeeNumber
- Loaded in 10,000-row chunks

3NF ETL Procedure

Step 3

Stored Procedure Normalization

- sp_etl_staging_to_oltp runs
- Normalizes flat table into 3NF
- Generates auto-increment PKs
- Populates departments & staff
- Populates projects & reviews

SCD2 Change Detection

Step 4

SCD Type 2 Versioning Engine

- Detects salary & role mutations
- Retires old row: is_current = 0
- Sets effective_end_date = NOW
- Inserts active row: is_current = 1
- Increments version counter

Dimensional Load

Step 5

Star Schema Data Warehouse

- Dim_Employee (SCD2 History)
- Dim_Department & Dim_Project
- Dim_Date (Role-Playing)
- Fact_PerformanceReviews
- Fact_ProjectAllocations

Direct SQL Window Queries & Star Schema Aggregations

Step 6: Streamlit Web UI, Business Dashboards & Advanced SQL Analytics

Executive KPI Hub

Real-time cards for total headcount, attrition rate, avg monthly salary, and project utilization rate.

Talent Leaderboards

SQL DENSE_RANK() queries top performers by department without rank gaps.

YoY Salary Progression

SQL LAG() and LEAD() calculate compensation delta between consecutive review cycles.

Burnout & Attrition

SQL NTILE(4) quartiles detect overtime vs low satisfaction flight risk employees.

Live SCD2 HR Forms

Streamlit forms for onboarding and department transfers triggering SCD2 in DW.